

Axiomatics 1 (only two axioms)

Damian Siniakowicz

July 22, 2026

Axiom 1

$$A \implies (B \implies A)$$

Axiom 2

$$(A \implies (B \implies C)) \implies ((A \implies B) \implies (A \implies C))$$

Theorem 1 (1.1)

$$(A \implies B) \implies (A \implies A)$$

$$1 : A \implies (B \implies A); \text{Axiom 1}$$

$$2 : (A \implies (B \implies A)) \implies ((A \implies B) \implies (A \implies A)); \text{Axiom 2}$$

$$3 : (A \rightarrow B) \rightarrow (A \rightarrow A); \text{MP 1, 2}$$

□

Theorem 2 (1.2)

$$A \rightarrow (B \rightarrow (C \rightarrow B))$$

$$1 : B \rightarrow (C \rightarrow B); \text{Axiom 1}$$

$$2 : (B \implies (C \implies B)) \rightarrow (A \rightarrow (B \rightarrow (C \rightarrow B))); \text{Axiom 1}$$

$$3 : A \rightarrow (B \rightarrow (C \rightarrow B)); \text{MP 1, 2}$$

□

Identity Theorem (2.1)

$$A \rightarrow A$$

$$1 : A \rightarrow (A \rightarrow A); \text{Axiom 1}$$

$$2 : A \rightarrow ((A \rightarrow A) \rightarrow A); \text{Axiom 1}$$

$$3 : (A \rightarrow ((A \rightarrow A) \rightarrow A)) \rightarrow ((A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)); \text{Axiom 2}$$

4 : $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$; MP 2, 3

5 : $A \rightarrow A$; MP 1, 4

□

Prefix Theorem (2.1)

$A \vdash B \implies A$

1 : $A \rightarrow (B \rightarrow A)$; Axiom 1

2 : $B \rightarrow A$; MP A, 1

□

Weakening Theorem (2.2)

$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow B))$

1 : $B \rightarrow (C \rightarrow B)$; Axiom 1

2 : $(A \rightarrow (B \rightarrow (C \rightarrow B))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow B)))$; Axiom 2

3 : $A \rightarrow (B \rightarrow (C \rightarrow B))$; Prefix A, 1

4 : $(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow B))$; MP 3, 2

□

Theorem 3 (2.3)

$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow B))$

1 : $(B \rightarrow C) \rightarrow (B \rightarrow B)$; Theorem 1

2 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow B))$; Prefix A, 1

□

Theorem 4 (2.4)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow C)))$

1 : $C \rightarrow (D \rightarrow C)$; Axiom 1

2 : $B \rightarrow (C \rightarrow (D \rightarrow C))$; Prefix B, 1

3 : $A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow C)))$; Prefix A, 2

□

Theorem 5 (2.5)

$$((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow A)$$

1 : $((A \Rightarrow B) \Rightarrow A) \Rightarrow ((A \Rightarrow B) \Rightarrow A)$; identity theorem

□

Theorem 6 (2.6)

$$(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow C))$$

1 : $(A \Rightarrow (B \Rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow C))$; identity theorem

□

Theorem 7 (3.1)

$$A \rightarrow (B \rightarrow B)$$

1 : $B \rightarrow B$; identity theorem

2 : $A \rightarrow (B \rightarrow B)$; prefix theorem A, 1

□

Theorem 8 (3.3)

$$A \rightarrow (B \rightarrow (C \rightarrow A))$$

1 : $A \rightarrow (C \rightarrow A)$; Axiom 1

2 : $(A \rightarrow (C \rightarrow A)) \rightarrow (A \rightarrow (B \rightarrow (C \rightarrow A)))$; Weakening 1, B

3 : $A \Rightarrow (B \Rightarrow (C \Rightarrow A))$; MP 1, 2

□

Simplification Theorem

$$B \vdash (A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)$$

1 : $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$; Axiom 2

2 : $((A \Rightarrow (B \Rightarrow C)) \Rightarrow ((A \Rightarrow B) \Rightarrow (A \Rightarrow C))) \rightarrow (((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow B)) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$

3 : $((A \Rightarrow (B \Rightarrow C)) \Rightarrow (A \Rightarrow B)) \Rightarrow ((A \Rightarrow (B \Rightarrow C)) \Rightarrow (A \Rightarrow C))$; MP 1, 2

4 : $A \rightarrow B$; Prefix: A, Assumption

5 : $((A \Rightarrow (B \Rightarrow C)) \Rightarrow (A \Rightarrow B))$; Prefix $(A \rightarrow (B \rightarrow C))$, 4

6 : $((A \implies (B \implies C)) \implies (A \implies C));$ MP 5, 3

□

Strengthening (3.4)

$(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow B)$

1 : $(A \rightarrow (A \rightarrow B)) \rightarrow ((A \rightarrow A) \rightarrow (A \rightarrow B));$ Axiom 2

2 : $A \rightarrow A;$ identity theorem

5 : $((A \rightarrow (A \rightarrow B)) \rightarrow ((A \rightarrow A) \rightarrow (A \rightarrow B))) \rightarrow ((A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow B));$ Simplification 1, 2

6 : $(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow B);$ MP 1, 5

□

Theorem 9 (3.5)

$((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B)$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow B);$ identity theorem

2 : $((A \rightarrow B) \rightarrow (A \rightarrow B)) \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B));$ Axiom 2

3 : $((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B);$ MP 1, 2

Theorem 10 (3.6)

$(A \rightarrow ((B \rightarrow A) \rightarrow C)) \rightarrow (A \rightarrow C)$

1 : $A \rightarrow (B \rightarrow A);$ Axiom 1

2 : $(A \rightarrow ((B \rightarrow A) \rightarrow C)) \rightarrow ((A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow C));$ Axiom 2

3 : $(A \rightarrow ((B \rightarrow A) \rightarrow C)) \rightarrow ((A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow C)) \rightarrow$

$(A \rightarrow ((B \rightarrow A) \rightarrow C)) \rightarrow (A \rightarrow C);$ Strengthening 1

4 : $(A \rightarrow ((B \rightarrow A) \rightarrow C)) \rightarrow (A \rightarrow C);$ MP 2, 3

Theorem 11 (3.7)

$(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow B))$

1 : $B \rightarrow B;$ Identity Theorem

2 : $A \rightarrow (B \rightarrow B);$ Prefix 1

3 : $(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow B));$ Prefix 2

Theorem 12 (3.2)

$$(A \rightarrow B) \rightarrow (A \rightarrow B)$$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow B)$; Identity Theorem

Trans 1 (3.8)

$$(A \rightarrow B) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B))$$

1 : $(C \rightarrow (A \rightarrow B)) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B))$; Axiom 2

2 : $(A \rightarrow B) \rightarrow ((C \rightarrow (A \rightarrow B)) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B)))$; Prefix 1

3 : $((A \rightarrow B) \rightarrow ((C \rightarrow (A \rightarrow B)) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B)))) \rightarrow$

$((A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow B))) \rightarrow ((A \rightarrow B) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B)))$; Axiom 2

4 : $((A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow B))) \rightarrow ((A \rightarrow B) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B)))$; MP 2, 3

5 : $(A \implies B) \implies (C \implies (A \implies B))$; Axiom 1

6 : $(A \rightarrow B) \rightarrow ((C \rightarrow A) \rightarrow (C \rightarrow B))$; MP 5, 4

Theorem 13 (3.9)

$$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow C)))$$

1 : $C \rightarrow (D \rightarrow C)$; Axiom 1

2 : $A \rightarrow (C \rightarrow (D \rightarrow C))$; Prefix 1

3 : $(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow C)))$; Prefix 1

Theorem 14 (3.10)

$$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow C)))$$

1 : $C \rightarrow (D \rightarrow C)$; Axiom 1

2 : $(C \rightarrow (D \rightarrow C)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow C)))$; Trans1

3 : $(B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow C))$; MP 1, 2

4 : $A \implies ((B \implies C) \implies (B \implies (D \implies C)))$; Prefix 3

Theorem 15 (3.11)

$$A \rightarrow (B \rightarrow ((C \rightarrow D) \rightarrow (C \rightarrow C)))$$

1 : $(C \implies D) \implies (C \implies C)$; Theorem_1

2 : $B \rightarrow ((C \rightarrow D) \rightarrow (C \rightarrow C))$; Prefix 1

3 : $A \implies (B \implies ((C \implies D) \implies (C \implies C)))$; Prefix 2

Theorem 16 (3.12)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow D))))$

1 : $D \rightarrow (E \rightarrow D)$; Axiom_1

2 : $A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow D))))$; Prefix; Prefix; Prefix 1

Theorem 17 (4.1)

$A \rightarrow (B \rightarrow (C \rightarrow C))$

1 : $B \rightarrow (C \rightarrow C)$; Theorem_7

2 : $A \rightarrow (B \rightarrow (C \rightarrow C))$; Prefix 1

Theorem 18 (4.2)

$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow A))$

1 : $A \rightarrow (C \rightarrow A)$; Axiom_1

2 : $(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow A))$; Prefix 1

Theorem 19 (4.3)

$(A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow A))$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow A)$; Theorem_1

2 : $((A \rightarrow B) \rightarrow (A \rightarrow A)) \rightarrow ((A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow A)))$; Weaken_Theorem

3 : $(A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow A))$; MP 1, 2

Theorem 20 (4.4)

$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow C))$

1 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow C))$; Theorem_7

Theorem 21 (4.5)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow B)))$

1 : $B \rightarrow (D \rightarrow B)$; Axiom_1

2 : $(B \rightarrow (D \rightarrow B)) \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow B)))$; Weaken_Theorem

3 : $B \implies (C \implies (D \implies B))$; MP 1, 2

4 : $A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow B)))$; Prefix_Theorem 3

Theorem 22 (4.6)

$(A \rightarrow (B \rightarrow C)) \rightarrow (D \rightarrow (E \rightarrow D))$

1 : $D \rightarrow (E \rightarrow D)$; Axiom_1

2 : $(A \rightarrow (B \rightarrow C)) \rightarrow (D \rightarrow (E \rightarrow D))$; Prefix_Theo 1

Theorem 23 (4.7)

$A \rightarrow ((B \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow C))$

1 : $(B \implies (B \implies C)) \implies (B \implies C)$; Strengthen_Theo

2 : $A \implies ((B \implies (B \implies C)) \implies (B \implies C))$; Prefix_Theo 1

Drop_True_Prefix

$A \vdash (A \rightarrow B) \rightarrow B$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow B)$; Identity_Theo

2 : $((A \implies B) \implies (A \implies B)) \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B))$; Axiom_2

3 : $((A \implies B) \implies A) \implies ((A \implies B) \implies B)$; MP 1, 2

4 : $(A \rightarrow B) \rightarrow A$; Prefix_Theo & Assumption

5 : $(A \implies B) \implies B$; MP 4, 3

Theorem 24 (5.1)

$((A \rightarrow (B \rightarrow A)) \rightarrow A) \rightarrow A$

1 : $A \rightarrow (B \rightarrow A)$; Axiom_1

2 : $((A \rightarrow (B \rightarrow A)) \rightarrow A) \rightarrow A$; Drop_True_Prefix 1

Theorem 25 (5.2)

$((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow C$

1 : $A \rightarrow (B \rightarrow A)$; Axiom_1

2 : $((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow C$; Drop_True_Prefix 1

Theorem 26 (5.3)

$$(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$$

1 : $A \rightarrow A$; Axiom_1

2 : $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$; Prefix_Theo 1

Theorem 27 (5.4)

$$(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow A)$$

1 : $A \rightarrow A$; Axiom_1

2 : $(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow A)$; Prefix_Theo 1

Theorem 28 (5.5)

$$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow C))$$

1 : $B \rightarrow (C \rightarrow C)$; Theorem_7

2 : $A \rightarrow (B \rightarrow (C \rightarrow C))$; Prefix_Theo

3 : $(A \rightarrow (B \rightarrow (C \rightarrow C))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow C)))$

4 : $(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow C))$; MP 2, 3

Theorem 29 (5.6)

$$(A \rightarrow B) \rightarrow (C \rightarrow (D \rightarrow C))$$

1 : $C \rightarrow (D \rightarrow C)$; Axiom_1

2 : $(A \rightarrow B) \rightarrow (C \rightarrow (D \rightarrow C))$; Prefix_1

Theorem 30 (5.7)

$$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow D)))$$

1 : $C \rightarrow (D \rightarrow D)$; Theo_7

2 : $B \rightarrow (C \rightarrow (D \rightarrow D))$; Prefix_Theo 1

3 : $A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow D)))$; Prefix_Theo 2

Theorem 31 (5.8)

$$((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow (D \rightarrow C)$$

1 : $A \rightarrow (B \rightarrow A)$; Axiom_1

2 : $((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow C$; Drop_True_Prefix 1

3 : $((((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow C) \rightarrow (((A \rightarrow (B \rightarrow A)) \rightarrow C) \rightarrow (D \rightarrow C)))$; Weaken_Th 2

4 : $((A \implies (B \implies A)) \implies C) \implies (D \implies C)$; MP 2, 3

Theorem 32 (5.9)

$(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow (A \rightarrow B))$

1 : $(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow (A \rightarrow B))$; Identity_Theo

Trans_Theo_2

$A \rightarrow B; B \rightarrow C \vdash A \rightarrow C$

1 : $A \rightarrow (B \rightarrow C)$; Prefix_ : Assumption

2 : $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$; Axiom_2

3 : $(A \rightarrow B) \rightarrow (A \rightarrow C)$; MP 1, 2

4 : $A \rightarrow C$; MP 0, 3

Theorem 33 (5.10)

$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow B)))$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow (D \rightarrow B))$; Weaken_Theo

2 : $(A \rightarrow (D \rightarrow B)) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow B)))$; Weaken_Theo

3 : $((A \rightarrow (D \rightarrow B)) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow B)))) \rightarrow (((A \rightarrow B) \rightarrow ((A \rightarrow (D \rightarrow B)))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow B)))))$; Tr
2, $A \rightarrow B$

4 : $((A \rightarrow B) \rightarrow ((A \rightarrow (D \rightarrow B)))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow B))))$; MP 2, 3

5 : $(A \implies B) \implies (A \implies (C \implies (D \implies B)))$; MP 1, 4

Theorem 34 (5.11)

$(A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow (D \rightarrow B)))$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow (D \rightarrow B))$; Weaken_Theo

2 : $((A \implies B) \implies (A \implies (D \implies B))) \rightarrow ((A \implies B) \implies (C \rightarrow (A \implies (D \implies B))))$; Weaken_Theo

3 : $(A \rightarrow B) \rightarrow (C \rightarrow (A \rightarrow (D \rightarrow B)))$; MP 1, 2

Theorem 35 (5.12)

$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow B)))$

1 : $B \rightarrow (D \rightarrow B)$; Axiom_1

2 : $(B \rightarrow C) \rightarrow (B \implies (D \implies B))$; Prefix_1

3 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow B)))$; Prefix_2

Theorem 36 (5.13)

$A \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (B \rightarrow B)))$

1 : $D \rightarrow (B \rightarrow B)$; Theorem_7

2 : $(B \rightarrow C) \rightarrow (D \implies (B \implies B))$; Prefix_1

3 : $A \rightarrow ((B \rightarrow C) \rightarrow (D \implies (B \implies B)))$; Prefix_2

Theorem 37 (5.14)

$A \rightarrow (B \rightarrow ((C \rightarrow D) \rightarrow (C \rightarrow D)))$

1 : $(C \rightarrow D) \rightarrow (C \rightarrow D)$; Identity_Theorem

2 : $B \rightarrow ((C \implies D) \implies (C \implies D))$; Prefix_1

3 : $A \rightarrow (B \rightarrow ((C \rightarrow D) \rightarrow (C \rightarrow D)))$; Prefix_2

Theorem 38 (5.15)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow C))))$

1 : $(C \rightarrow (D \rightarrow C)) \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow C)))$; Weaken_Theo

2 : $C \rightarrow (D \rightarrow C)$; Axiom_1

3 : $C \rightarrow (D \rightarrow (E \rightarrow C))$; MP 2,1

4 : $B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow C)))$; Prefix_3

5 : $A \rightarrow (B \implies (C \implies (D \implies (E \implies C))))$; Prefix_4

Theorem 39 (6.1)

$(A \rightarrow A) \rightarrow (A \rightarrow A)$

1 : $(A \rightarrow A) \rightarrow (A \rightarrow A)$; Identity_Theorem

Condensed_Mopo (6.2)

$A \rightarrow ((A \rightarrow B) \rightarrow B)$

1 : $(A \rightarrow B) \rightarrow (A \rightarrow B)$; Identity_Theo

2 : $((A \Rightarrow B) \Rightarrow (A \Rightarrow B)) \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B))$; Axiom_2

3 : $((A \Rightarrow B) \Rightarrow A) \Rightarrow ((A \Rightarrow B) \Rightarrow B)$; MP 1, 2

4 : $A \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow ((A \rightarrow B) \rightarrow B))$; Prefix_3

5 : $(A \Rightarrow (((A \Rightarrow B) \Rightarrow A) \Rightarrow ((A \Rightarrow B) \Rightarrow B))) \rightarrow ((A \rightarrow ((A \rightarrow B) \rightarrow A)) \rightarrow (A \rightarrow ((A \rightarrow B) \rightarrow B)))$; Axiom_2

6 : $(A \rightarrow ((A \rightarrow B) \rightarrow A)) \rightarrow (A \rightarrow ((A \rightarrow B) \rightarrow B))$; MP 4, 5

7 : $A \rightarrow ((A \rightarrow B) \rightarrow A)$; Axiom_1

8 : $A \Rightarrow ((A \Rightarrow B) \Rightarrow B)$; MP 6, 7

Theorem 40 (6.3)

$((A \rightarrow A) \rightarrow B) \rightarrow (A \rightarrow A)$

1 : $A \rightarrow A$; Identity_Theorem

2 : $((A \Rightarrow A) \Rightarrow B) \Rightarrow (A \Rightarrow A)$; Prefix_1

Transitivity Theorem 3

$A \rightarrow B \vdash (B \rightarrow C) \rightarrow (A \rightarrow C)$

1 : $(B \rightarrow C) \rightarrow (A \rightarrow (B \rightarrow C))$; Axiom_1

2 : $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$; Axiom_2

3 : $((A \rightarrow B) \rightarrow (A \rightarrow C)) \rightarrow (A \rightarrow C)$; Drop_True_Prefix 0

4 : $(B \rightarrow C) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$; Transitivity_Theo 1, 2

5 : $(B \rightarrow C) \rightarrow (A \rightarrow C)$; Transitivity_Theo 4, 3

Theorem 41 (6.4)

$((A \rightarrow B) \rightarrow C) \rightarrow (B \rightarrow C)$

1 : $B \rightarrow (A \rightarrow B)$; Axiom_1

2 : $((A \Rightarrow B) \Rightarrow C) \Rightarrow ((A \rightarrow B) \rightarrow C)$; Identity_Theo

3 : $((((A \Rightarrow B) \Rightarrow C) \Rightarrow ((A \rightarrow B) \rightarrow C)) \rightarrow (((A \rightarrow B) \rightarrow C) \rightarrow (B \rightarrow ((A \Rightarrow B) \Rightarrow C))))$; Weaken_Theo

4 : $((A \rightarrow B) \rightarrow C) \rightarrow (B \rightarrow ((A \Rightarrow B) \Rightarrow C))$; MP 2, 3

5 : $(B \rightarrow ((A \rightarrow B) \rightarrow C)) \rightarrow ((B \rightarrow (A \rightarrow B)) \rightarrow (B \rightarrow C))$; Axiom_2

6 : $((A \rightarrow B) \rightarrow C) \rightarrow ((B \rightarrow (A \rightarrow B)) \rightarrow (B \rightarrow C))$; Trans_Theo 4, 5

7 : $((B \implies (A \implies B)) \implies (B \implies C)) \rightarrow (B \rightarrow C)$; Drop_True_Prefix 1

8 : $((A \rightarrow B) \rightarrow C) \rightarrow (B \rightarrow C)$; Trans_Theo 6, 7

Theorem 42 (6.5)

$(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow A)$

1 : $A \rightarrow A$; Identity_Theo

2 : $(A \implies (B \implies C)) \implies (A \implies A)$; Prefix_Theo 1

Theorem 43 (6.6)

$(A \rightarrow (B \rightarrow C)) \rightarrow (D \rightarrow D)$

1 : $D \rightarrow D$; Identity_Theo

2 : $(A \implies (B \implies C)) \implies (D \implies D)$; Prefix_Theo 1

Theorem 44 (6.7)

$A \rightarrow ((A \rightarrow B) \rightarrow (C \rightarrow B))$

1 : $A \rightarrow ((A \rightarrow B) \rightarrow B)$; condensed_mopo

2 : $((A \rightarrow B) \rightarrow B) \rightarrow ((A \rightarrow B) \rightarrow (C \rightarrow B))$; weaken_theo

3 : $A \implies ((A \implies B) \implies (C \implies B))$; Trans_Theo 1, 2

Theorem 45 (6.8)

$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow A))$

1 : $A \rightarrow (C \rightarrow A)$; Axiom_1

2 : $(A \rightarrow (C \rightarrow A)) \rightarrow (A \rightarrow (B \rightarrow (C \rightarrow A)))$; Weaken_Theo

3 : $A \rightarrow (B \rightarrow (C \rightarrow A))$; MP 1, 2

4 : $(B \rightarrow (C \rightarrow A)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow A))$; Axiom_2

5 : $A \implies ((B \implies C) \implies (B \implies A))$; Trans_Theo 3, 4

Theorem 46 (6.9)

$A \rightarrow (B \rightarrow ((B \rightarrow C) \rightarrow B))$

1 : $B \rightarrow ((B \rightarrow C) \rightarrow B)$; Axiom_1

2 : $A \implies (B \implies ((B \implies C) \implies B))$; Prefix_1

Theorem 47 (6.10)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow A)))$

1 : $A \rightarrow (D \rightarrow A)$; Axiom_1

2 : $(A \implies (D \implies A)) \rightarrow (A \implies (C \rightarrow (D \implies A)))$; Weaken_Theo

3 : $A \rightarrow (C \implies (D \rightarrow A))$; MP 1, 2

4 : $(A \implies (C \rightarrow (D \implies A))) \rightarrow (A \rightarrow (B \rightarrow (C \implies (D \rightarrow A))))$; Weaken_Theo

5 : $A \implies (B \implies (C \rightarrow (D \implies A)))$; MP 3, 4

Theorem 48 (6.11)

$((A \rightarrow B) \rightarrow (A \rightarrow A)) \rightarrow C \rightarrow C$

1 : $(A \implies B) \implies (A \implies A)$; Theo_1

2 : $((A \implies B) \implies (A \implies A)) \implies C \implies C$; Drop_True_Prefix 1

Theorem 49 (6.12)

$((A \rightarrow (B \rightarrow A)) \rightarrow A) \rightarrow (B \rightarrow A)$

1 : $A \rightarrow (B \rightarrow A)$; Axiom_1

2 : $((A \implies (B \implies A)) \implies A) \implies A$; Drop_True_Prefix 1

3 : $((A \rightarrow (B \rightarrow A)) \rightarrow A) \rightarrow A \rightarrow ((A \implies (B \implies A)) \implies A) \implies (B \rightarrow A)$; Weaken_Theo

4 : $((A \rightarrow (B \rightarrow A)) \rightarrow A) \rightarrow (B \rightarrow A)$; MP 2, 3

Theorem 50 (6.13)

$((A \rightarrow (B \rightarrow (C \rightarrow B))) \rightarrow A) \rightarrow A$

1 : $B \rightarrow (C \rightarrow B)$; Axiom_1

2 : $A \rightarrow (B \rightarrow (C \rightarrow B))$; Prefix_1

3 : $((A \rightarrow (B \rightarrow (C \rightarrow B))) \rightarrow A) \rightarrow A$; Drop_True_Prefix 2

Theorem 51 (6.14)

$((A \rightarrow (B \rightarrow (C \rightarrow B))) \rightarrow D) \rightarrow D$

1 : $B \rightarrow (C \rightarrow B)$; Axiom_1

2 : $A \rightarrow (B \rightarrow (C \rightarrow B))$; Prefix_1

3 : $((A \rightarrow (B \rightarrow (C \rightarrow B))) \rightarrow D) \rightarrow D$; Drop_True_Prefix 2

Theorem 52 (6.15)

$((A \rightarrow B) \rightarrow A) \rightarrow (C \rightarrow (D \rightarrow C))$

1 : $C \rightarrow (D \rightarrow C)$; Axiom_1

2 : $((A \Rightarrow B) \Rightarrow A) \Rightarrow (C \Rightarrow (D \Rightarrow C))$; Prefix_1

Theorem 53 (6.16)

$((A \rightarrow B) \rightarrow C) \rightarrow ((A \rightarrow B) \rightarrow C)$

1 : $((A \Rightarrow B) \Rightarrow C) \Rightarrow ((A \Rightarrow B) \Rightarrow C)$; Identity_Theo

Drop_True_Prefix_2

$B \vdash (A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)$

1 : $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$; Axiom_2

2 : $A \rightarrow B$; Prefix_0

3 : $((A \rightarrow B) \rightarrow (A \rightarrow C)) \rightarrow (A \rightarrow C)$; Drop_True_Prefix

4 : $(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)$; Trans_Theo 1, 3

Theorem 54 (6.17)

$(A \rightarrow ((A \rightarrow A) \rightarrow B)) \rightarrow (A \rightarrow B)$

1 : $A \rightarrow A$; Identity_Theo

2 : $(A \rightarrow ((A \rightarrow A) \rightarrow B)) \rightarrow (A \rightarrow B)$; Drop_True_Prefix_2 1

Theorem 55 (6.18)

$(A \rightarrow ((B \rightarrow B) \rightarrow C)) \rightarrow (A \rightarrow C)$

1 : $B \rightarrow B$; Identity_Theo

2 : $(A \Rightarrow ((B \Rightarrow B) \Rightarrow C)) \Rightarrow (A \Rightarrow C)$; Drop_True_Prefix_2 1

Theorem 56 (6.19)

$(A \rightarrow (A \rightarrow B)) \rightarrow (C \rightarrow (A \rightarrow B))$

1 : $(A \rightarrow (A \rightarrow B)) \rightarrow (A \rightarrow B)$; strengthen_theo

Theorem 57 (6.20)

$$(A \rightarrow (B \rightarrow (C \rightarrow D))) \rightarrow (A \rightarrow A)$$

1 : $A \rightarrow A$; Identity_Theo

2 : $(A \implies (B \implies (C \implies D))) \implies (A \implies A)$; Prefix_Theo 1

Theorem 58 (6.21)

$$(A \rightarrow B) \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow A)$$

1 : $(A \rightarrow B) \rightarrow (((A \rightarrow B) \rightarrow A) \rightarrow A)$; Condensed_Mopo

Theorem 59 (6.22)

$$(A \rightarrow B) \rightarrow ((C \rightarrow D) \rightarrow (C \rightarrow C))$$

1 : $(C \implies D) \implies (C \implies C)$; Theo_1

2 : $(A \implies B) \implies ((C \implies D) \implies (C \implies C))$; Prefix_Theo 1

Theorem 60 (6.23)

$$(A \rightarrow B) \rightarrow (A \rightarrow (C \rightarrow (D \rightarrow D)))$$

1 : $C \rightarrow (D \rightarrow D)$; Theo_7

2 : $A \implies (C \implies (D \implies D))$; Prefix_Theo 1

3 : $(A \implies B) \implies (A \implies (C \implies (D \implies D)))$; Prefix_Theo 2

Theorem 61 (6.24)

$$(A \rightarrow B) \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow D)))$$

1 : $D \rightarrow (E \rightarrow D)$; Axiom_1

2 : $C \rightarrow (D \rightarrow (E \rightarrow D))$; Prefix_1

3 : $(A \rightarrow B) \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow D)))$; Prefix_2

Theorem 62 (6.25)

$$A \rightarrow (((B \rightarrow (C \rightarrow B)) \rightarrow B) \rightarrow B)$$

1 : $B \rightarrow (C \rightarrow B)$; Axiom_1

2 : $((B \implies (C \implies B)) \implies B) \implies B$; Drop_True_Prefix 1

3 : $A \rightarrow (((B \rightarrow (C \rightarrow B)) \rightarrow B) \rightarrow B)$; Prefix_2

Theorem 63 (6.26)

$A \rightarrow (((B \rightarrow (C \rightarrow B)) \rightarrow D) \rightarrow D)$

1 : $B \rightarrow (C \rightarrow B)$; Axiom_1

2 : $((B \implies (C \implies B)) \implies D) \implies D$; Drop_True_Prefix 1

3 : $A \rightarrow (((B \rightarrow (C \rightarrow B)) \rightarrow D) \rightarrow D)$; Prefix_2

Theorem 64 (6.27)

$A \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow C))$

1 : $A \implies ((A \implies (B \implies C)) \implies (B \implies C))$; Condensed_Mopo

Theorem 65 (6.28)

$A \rightarrow ((B \rightarrow (B \rightarrow B)) \rightarrow (B \rightarrow B))$

1 : $(B \implies (B \implies B)) \implies (B \implies B)$; Strengthen_Theo

2 : $A \rightarrow ((B \rightarrow (B \rightarrow B)) \rightarrow (B \rightarrow B))$; Prefix_Theo 1

Theorem 66 (6.29)

$A \rightarrow ((B \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow B))$

1 : $B \rightarrow B$; Identity_Theo

2 : $(B \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow B)$; Prefix_1

3 : $A \implies ((B \implies (B \implies C)) \implies (B \implies B))$; Prefix_2

Theorem 67 (6.30)

$A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow D)))$

1 : $C \rightarrow (D \rightarrow D)$; Theo_7

2 : $B \rightarrow (C \implies (D \implies D))$; Prefix_1

3 : $(B \rightarrow (C \implies (D \implies D))) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow D)))$; Axiom_2

4 : $(B \implies C) \implies (B \implies (D \implies D))$; MP 2, 3

5 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (D \rightarrow D)))$; Prefix_4

Theorem 68 (6.31)

$A \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (B \rightarrow C)))$

1 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow C)); \text{Theo_7}$

2 : $(B \implies C) \implies (B \implies C); \text{Identity_Theo}$

3 : $((B \rightarrow C) \rightarrow (B \rightarrow C)) \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (B \rightarrow C))); \text{Weaken_Theo}$

4 : $A \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (B \rightarrow C))); \text{Trans_1, 3}$

Theorem 69 (6.32)

$A \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (E \rightarrow D)))$

1 : $D \rightarrow (E \rightarrow D); \text{Axiom_1}$

2 : $(B \rightarrow C) \rightarrow (D \rightarrow (E \rightarrow D)); \text{Prefix_1}$

3 : $A \rightarrow ((B \rightarrow C) \rightarrow (D \rightarrow (E \rightarrow D))); \text{Prefix_2}$

Theorem 70 (6.33)

$A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow E))))$

1 : $D \rightarrow (E \rightarrow E); \text{Theo_7}$

2 : $C \rightarrow (D \rightarrow (E \rightarrow E)); \text{Prefix_1}$

3 : $B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow E))); \text{Prefix_2}$

4 : $A \rightarrow (B \rightarrow (C \rightarrow (D \rightarrow (E \rightarrow E)))); \text{Prefix_3}$

Theorem 71 (7.1)

$((A \rightarrow A) \rightarrow A) \rightarrow A$

1 : $A \rightarrow A; \text{Identity_Theo}$

2 : $((A \implies A) \implies A) \implies A; \text{Drop_True_Prefix 1}$

Theorem 72 (7.2)

$((A \rightarrow A) \rightarrow B) \rightarrow B$

1 : $A \rightarrow A; \text{Identity_Theo}$

2 : $((A \implies A) \implies B) \implies B; \text{Drop_True_Prefix 1}$

Theorem 73 (7.3)

$$(A \rightarrow B) \rightarrow (C \rightarrow C)$$

1 : $(A \rightarrow B) \rightarrow (C \rightarrow C)$; Theo_7

Theorem 74 (7.4)

$$A \rightarrow (A \rightarrow (B \rightarrow A))$$

1 : $(A \rightarrow (B \rightarrow A))$; Axiom_1

2 : $A \rightarrow (A \rightarrow (B \rightarrow A))$; Prefix_1

Theorem 75 (7.5)

$$A \rightarrow (B \rightarrow (B \rightarrow B))$$

1 : $B \rightarrow (B \rightarrow B)$; Axiom_1

2 : $A \Rightarrow (B \Rightarrow (B \Rightarrow B))$; Prefix_1

Theorem 76 (7.6)

$$((A \rightarrow (A \rightarrow A)) \rightarrow A) \rightarrow A$$

1 : $A \rightarrow (A \rightarrow A)$; Axiom_1

2 : $((A \Rightarrow (A \Rightarrow A)) \Rightarrow A) \Rightarrow A$; Drop_True_Prefix 1

Theorem 77 (7.7)

$$((A \rightarrow A) \rightarrow B) \rightarrow (C \rightarrow B)$$

1 : $A \rightarrow A$; Identity_Theo

2 : $((A \rightarrow A) \rightarrow B) \rightarrow B$; Drop_True_Prefix

3 : $((A \rightarrow A) \rightarrow B) \rightarrow ((A \rightarrow A) \rightarrow B) \rightarrow (C \rightarrow B)$; Weaken_Theo

4 : $((A \Rightarrow A) \Rightarrow B) \Rightarrow (C \Rightarrow B)$; MP 2, 3

Theorem 78 (7.8)

$$(A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow A)$$

1 : $A \rightarrow A$; Identity_Theo

2 : $(A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow A)$; Prefix_1

Suffix Transs Theo 1

$B \rightarrow C; A \rightarrow (C \rightarrow D) \vdash A \rightarrow (B \rightarrow D)$

1 : $(C \rightarrow D) \rightarrow (B \rightarrow (C \rightarrow D));$ Axiom_1

2 : $(B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D));$ Axiom_2

3 : $(C \rightarrow D) \rightarrow ((B \implies C) \implies (B \implies D));$ Trans_Theo 1, 2

4 : $(C \rightarrow D) \rightarrow (B \rightarrow D);$ Drop_2nd_Prefix 3

5 : $A \rightarrow ((C \rightarrow D) \rightarrow (B \rightarrow D));$ Prefix_4

6 : $(A \rightarrow ((C \rightarrow D) \rightarrow (B \rightarrow D))) \rightarrow ((A \rightarrow (C \rightarrow D)) \rightarrow (A \rightarrow (B \rightarrow D)));$ Axiom_2

7 : $(A \rightarrow (C \rightarrow D)) \rightarrow (A \rightarrow (B \rightarrow D));$ MP 5, 6

8 : $A \rightarrow (B \rightarrow D);$ MP 0, 7

Suffix Trans Theo 2

$A \rightarrow (B \rightarrow C); C \rightarrow D \vdash A \rightarrow (B \rightarrow D)$

1 : $B \rightarrow (C \rightarrow D);$ Prefix_0

2 : $(B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D));$ Axiom_2

3 : $(B \rightarrow C) \rightarrow (B \rightarrow D);$ MP 1, 2

4 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D));$ Prefix_3

5 : $(A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D))) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow D)));$ Axiom_2

6 : $((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow D)));$ MP 4, 5

7 : $A \implies (B \implies D);$ MP 0, 6

Permutation

$(A \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow (A \rightarrow C))$

1 : $(B \rightarrow C) \rightarrow (B \rightarrow (A \rightarrow C));$ Weaken_Theo

2 : $A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow (A \rightarrow C)));$ Prefix_1

3 : $(A \implies ((B \implies C) \implies (B \implies (A \implies C)))) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow (A \rightarrow C))));$
Axiom_2

4 : $(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow (A \rightarrow C)));$ PM 2, 3

5 : $(A \rightarrow (B \rightarrow (A \rightarrow C))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow (A \rightarrow C)));$ Axiom_2

6 : $(A \rightarrow (A \rightarrow C)) \rightarrow (A \rightarrow C);$ Strengthen_Theo

7 : $(A \rightarrow (B \rightarrow (A \rightarrow C))) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C));$ Suffix_Trans_2; 5, 6

8 : $((A \rightarrow B) \rightarrow (A \rightarrow C)) \rightarrow (B \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C)));$ Axiom_1

9 : $(B \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))) \rightarrow ((B \rightarrow (A \rightarrow B)) \rightarrow (B \rightarrow (A \rightarrow C)))$; Axiom_2

10 : $B \rightarrow (A \rightarrow B)$; Axiom_1

11 : $((B \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))) \rightarrow ((B \rightarrow (A \rightarrow B)) \rightarrow (B \rightarrow (A \rightarrow C)))) \rightarrow ((B \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))) \rightarrow (B \rightarrow (A \rightarrow C)))$; Drop_True_Prefix_2; 10

12 : $(B \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))) \rightarrow (B \rightarrow (A \rightarrow C))$; MP 9, 11

Suffix_Trans

$A \rightarrow (B \rightarrow C); A \rightarrow (C \rightarrow D) \vdash A \rightarrow (B \rightarrow D)$

1 : $(B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D))$; Axiom_2

2 : $A \rightarrow ((B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D)))$; Prefix_1

3 : $(A \rightarrow ((B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D)))) \rightarrow (A \rightarrow (B \rightarrow (C \rightarrow D))) \rightarrow (A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D)))$; Axiom_2

4 : $(A \rightarrow (B \rightarrow (C \rightarrow D))) \rightarrow (A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D)))$; MP 2, 3

5 : $(A \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D))) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow D)))$; Axiom_2

6 : $(A \rightarrow (B \rightarrow (C \rightarrow D))) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (B \rightarrow D)))$; Transs_Theo 4, 5

7 : $(A \rightarrow (B \rightarrow (C \rightarrow D))) \rightarrow (A \rightarrow (B \rightarrow D))$; Drop_True_Prefix_2; 0, 6

8 : $(A \rightarrow (C \rightarrow D)) \rightarrow (A \rightarrow (B \rightarrow (C \rightarrow D)))$; Weaken_Theo

9 : $A \rightarrow (B \rightarrow (C \rightarrow D))$; MP 0, 8

10 : $A \rightarrow (B \rightarrow D)$; MP 9, 7

Theorem 79(7.9)

$A \rightarrow ((A \rightarrow (A \rightarrow B)) \rightarrow B)$